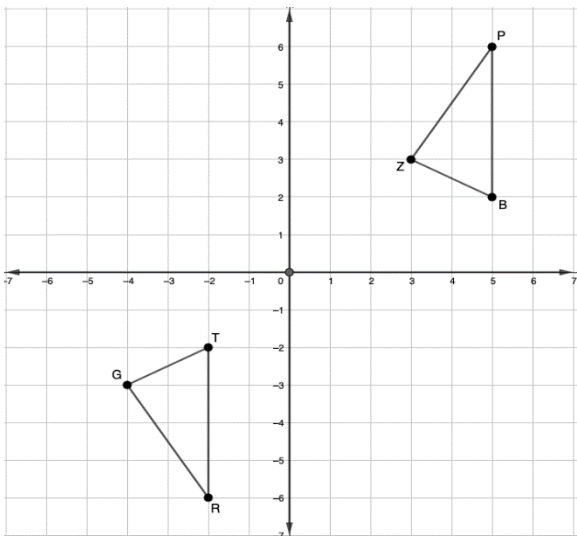


Geometry
Unit 4
Lesson 8 Homework

Name Answer Key
Date _____
Period _____

For each coordinate grid with the pre-image of $\triangle TGR$ and image $\triangle BZP$, where $\triangle TGR \cong \triangle BZP$. Complete the table with the missing written and algebraic description(s) that map $\triangle TGR$ onto $\triangle BZP$.

Coordinate Grid	Written Description	Algebraic Description
	1. 180° rotation about the origin.	2. $(x, y) \rightarrow (-x, -y)$
	3. Translate right 4 units and up 3 units, then rotate 90° counterclockwise about the origin.	4. $(x, y) \rightarrow (x + 4, y + 3)$ $(x, y) \rightarrow (-y, x)$
	5. Reflect over the line $y = x$, then translate right 2 units and down 2 units.	6. $(x, y) \rightarrow (y, x)$ $(x, y) \rightarrow (x + 2, y - 2)$



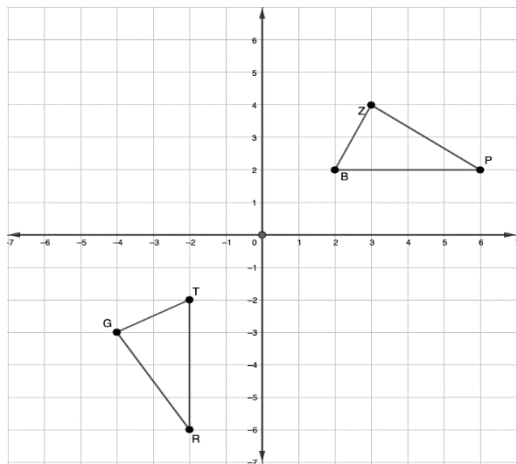
7.

Translate right 7 units,
then reflect over the
 x -axis.

8.

$$(x, y) \rightarrow (x + 7, y)$$

$$(x, y) \rightarrow (x, -y)$$



9.

Rotate 90° clockwise,
then reflect over the
 y -axis.

10.

$$(x, y) \rightarrow (y, -x)$$

$$(x, y) \rightarrow (-x, y)$$